RedCat 51 II Flat Panel — Bill of Materials

Slip-on flat-field panel for the William Optics RedCat 51 II · three 3D-printed parts + a printed 88 mm diffuser + an LED ring · Amazon links shortened to the plain `/dp/ASIN` form

Required

ITEM	COST	LINK
BNTECHGO 22 AWG PVC solid hook-up wire, red + black, 100 ft each	$14.48	amazon.com/dp/B07JNB712X
SUYOOULIN LED strip, SMD 2835, 12 V, 1200 LEDs, 16.4 ft, natural white 4500 K, non-waterproof	$14.99	amazon.com/dp/B0CGLPD6FJ
MP1584EN DC–DC buck converter, 8-pack 1 used. Its trim pot sets brightness once, during assembly.	$7.49	amazon.com/dp/B0H2VPZ6QR
12 V 5 A DC barrel-plug pigtails, 2.1 × 5.5 mm, male + female, 5 pairs	$5.98	amazon.com/dp/B0D8BDX6Q5
ELEGOO Rapid PLA+ filament bundle, 1.75 mm, black + white, 2 kg Back wall must be **white** (or white on its top face).	$22.80	amazon.com/dp/B0C6XDTHYN
M3 socket-head screw / nut kit, 461 pcs Need **3 × M3×40 mm bolts + 3 hex nuts** (or M3×35 with the optional nut-trap collar).	$11.99	amazon.com/dp/B0FGJ9FRGQ
Diffuser disc, 88 mm — **3D-printed** (0.6 or 1.0 mm, white) Printed from the filament above. No extra purchase.	—	printed
Subtotal (priced items)	**$77.73**	

Optional the panel builds fine without these

ITEM	COST	LINK
88 mm × 3 mm opal / milky acrylic disc, instead of the printed diffuser Laser-cut from `diffuser_88mm.dxf`.	—	no link; laser-cut service
20 pcs 10 mm 2-pin COB LED strip connectors, strip-to-wire, solderless Or solder wires straight to the strip.	$8.99	amazon.com/dp/B0C58Q6GKF
M3 heat-set threaded inserts (Ruthex) Not needed. The build uses hex nuts.	$9.99	amazon.com/dp/B08BCRZZS3
Subtotal	**$18.98**	

Tools & misc you may already own these

ITEM	COST	NOTES
Soldering iron	$30.00	cheaper irons available
Soldering flux	$7.00	optional; helps tin adhere to components
Heat-shrink tubing	$9.00	kit with various sizes
Solder	$7.00	get rosin core
Subtotal	**$53.00**	

Prices are from Oct 2026 and will drift. If a link is dead, search the item name. Assembly guide: project README.